

Date:

Roll No.

III SEMESTER B.Tech (ECE, EVDT, EIOT) END-SEMESTER EXAMINATION NOVEMBER 2025

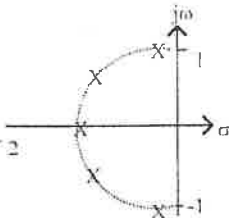
Course Code- ECECC302 / VTBECC302 / EIECC302 / ECECC05 / EIECC05
Course Title- Signals & Systems
Time: 3:00 Hrs

Max Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q.N.	Questions	Marks	CO
Q1	Attempt any 2 out of 3 parts		
	Let two functions be defined by		
a	i $x(t) = \begin{cases} 1, & \sin(20\pi t) \geq 0 \\ -1, & \sin(20\pi t) < 0 \end{cases}$ and ii $x_2(t) = \begin{cases} t, & \sin(20\pi t) \geq 0 \\ -t, & \sin(20\pi t) < 0 \end{cases}$ Plot these two functions. Are they periodic? If yes, what are the time periods? Evaluate the following expressions: (i) $\int_{-\infty}^{\infty} \left[\cos \frac{\pi}{2} (t-5) \right] \cdot \delta(2t-3) dt$ (ii) $e^{2t} \cos \left(\frac{50}{\pi} t \right) \delta(t+1)$	05	
b	(iii) Compute the Energy/Power of $x[n] = \cos \left[\frac{\pi n}{4} \right] + \sin \left[\frac{\pi n}{8} \right] - 2 \cos \left[\frac{\pi n}{2} \right]$	05	CO1
c	Consider a discrete-time system with input $x[n]$ and $y[n]$ related by $y[n] = \sum_{k=-n_0}^{n+n_0} x[k]$. Where n_0 is a finite positive integer. i Is this system linear? ii Is this system time-invariant? iii Is this system causal? iv Is this system stable?	05	
Q2	Attempt any 2 out of 3 parts		
a	Which continuous time signal with fundamental time period 'T', will have the following Fourier series coefficients i $X_k = \frac{\sin^2(k\pi/2)}{4(k\pi/2)}$ ii $X_k = 2$ Plot the signal.	05	CO2

	If $f(t) * g(t) = c(t)$, then express i $f(at) * g(at)$ ii $f(-t) * g(-t)$ iii $f(t-t_0) * g(t+t_1)$ In terms of $c(t)$.	05	
Q3	Attempt any 2 out of 3 parts		
b	Following information are given about a periodic signal $x[n]$ with fundamental period $N = 8$ and Fourier series coefficient X_k : i $X_k = -X_{k-4}$ ii $y[n] = \left(\frac{(1)^n + (-1)^n}{2} \right) x[n-1]$ Find the Fourier series Coefficients of $y(n)$.	05	
a	Find the following numerical values. i $X(j\omega)_{j\omega=1/2}$ when $x(t) = \text{rect}(t) * \text{rect}(2t)$ ii $x(0)$, when $X(j\omega) = 2 \text{sinc}(\omega/2\pi) * 3 \text{sinc}(\omega/\pi)$ Find an expression for $y(t)$ using Fourier Transform. Do not use convolution.	05	
b	i $y(t) = e^{-t} u(t) * \cos(\pi t)$ ii $y(t) = \text{sinc}(t) * \text{sinc}^2(t/2)$ Suppose we want to design a discrete-time LTI system which has the property that if the input is $x[n] = \left(\frac{1}{2} \right)^n u[n] - \frac{1}{4} \left(\frac{1}{2} \right)^{n-1} u[n-1]$ then the output is $y[n] = \left(\frac{1}{3} \right)^n u[n]$	05	CO3
c	(i) Find the impulse response and frequency response of a discrete time LTI system that has the foregoing property (ii) Is the system stable, causal.	05	
Q4	Attempt any 4 out of 6 parts		
a	For the below given pole plot of $X(s)$, i. Determine $x(t)$ if $x(t)e^{-3t}$ is absolutely integrable. ii. Determine $x(t)$ if $x(t) * (e^{-t} u(t))$ is absolutely integrable (Here * means convolution).	2.5+ 2.5	CO4

	b	We are given the following five facts about a real signal $x(t)$ with Laplace transform $X(s)$: 1. $X(s)$ has exactly two poles. 2. $X(s)$ has no zeros in the finite s-plane. 3. $X(s)$ has a pole at $s = -1 + j$. 4. $e^{2t}x(t)$ is not absolutely integrable. 5. $X(0) = 8$ Determine $X(s)$ and Compute $x(t)$.	3+2	
	c	$X(s)$ has 5 poles at $s = -1$, $s = \sqrt{\frac{1}{2}} \pm j\sqrt{\frac{1}{2}}$, $s = \pm j$ - i Is the signal $x(t)$ real, even, oscillatory? - ii Find all the possible two sided functions $x(t)$. 	02+ 03	
Q5	Attempt any 2 out of 3 parts			CO5
	a	State and prove all the properties of ROC of Z Transform.	05	
	b	Find the inverse z-transform of this function in closed form using partial-fraction expansion, - i $X(z) = \frac{z^2}{z^2 - z + 1/4}$, $z < 0.5$ - ii $X(z) = \frac{1}{z^2 + 0.8z + 0.16}$, $z > 0.4$	05	
	c	The transfer function of a system is given as $H(z) = \frac{z + 0.5}{(z + 0.4)(z - 2)}$ Specify the ROC of $H(z)$ and determine $h(n)$ for the following conditions: (i) The system is causal. (ii) The system is stable (iii) Can the given system be both causal and stable?	05	